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Palaeomagnetism and radiometric dating of the British Tertiary Igneous Province: Muck and Eigg

P. Dagley and A. E. Mussett *Sub-Department of Geophysics, Oliver Lodge Laboratory, The University, PO Box 147, Liverpool L69 3PX*

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Summary. The lavas, dykes and other intrusive bodies of Muck and Eigg have been sampled for palaeomagnetic purposes. All the bodies are reversely magnetized except for four dykes of Muck which are normally magnetized. The lavas show variations of direction of magnetism believed to be a record of secular variation.

The radiometric dates support the geological evidence that the lavas of Eigg and Muck predate parts of the Rhum central igneous complex, the lavas of Rhum and Canna and some of those of Skye. The combined evidence shows that most of the activity took place during an R-N-R-N sequence between 63 and 58 Ma. The younger R-N sequence is the same as the first part of the R-N-R sequence found in Mull but the Sgurr pitchstone of Eigg and parts of the Eastern Redhills of Skye belong to later reversed intervals and correspond to Chron 26r and Chron 26. The igneous activity occurred over a 10 Myr period from about 63 Ma and can probably be associated with Chrons 27r to 23r of the Harland *et al.* time-scale.

Key words: dating, igneous, palaeomagnetism, Scotland, stratigraphy, Tertiary

Introduction

The work described in this paper complements that reported for Rhum and Canna (Dagley & Mussett 1981). The aims of this investigation were to relate the igneous activity of each of the Small Isles to one another, to that of their near neighbour, Skye, and to the rest of the British Tertiary Igneous Province (BTIP). Broadly this can be achieved by combining palaeomagnetic polarity with radiometric age and stratigraphic links, aided in some cases by correlation of directions of magnetization between lava profiles at different locations on one island, as was done by Khan (1960) and Wilson, Dagley & Ade-Hall (1972b) for Skye and Wilson (1970) for northern Ireland. Examination of magnetic properties, particularly thermomagnetic analysis which by the shape of the curve and the Curie point can provide evidence of composition and history (Ade-Hall, Palmer & Hubbard 1971), may also be a help in making such correlations.

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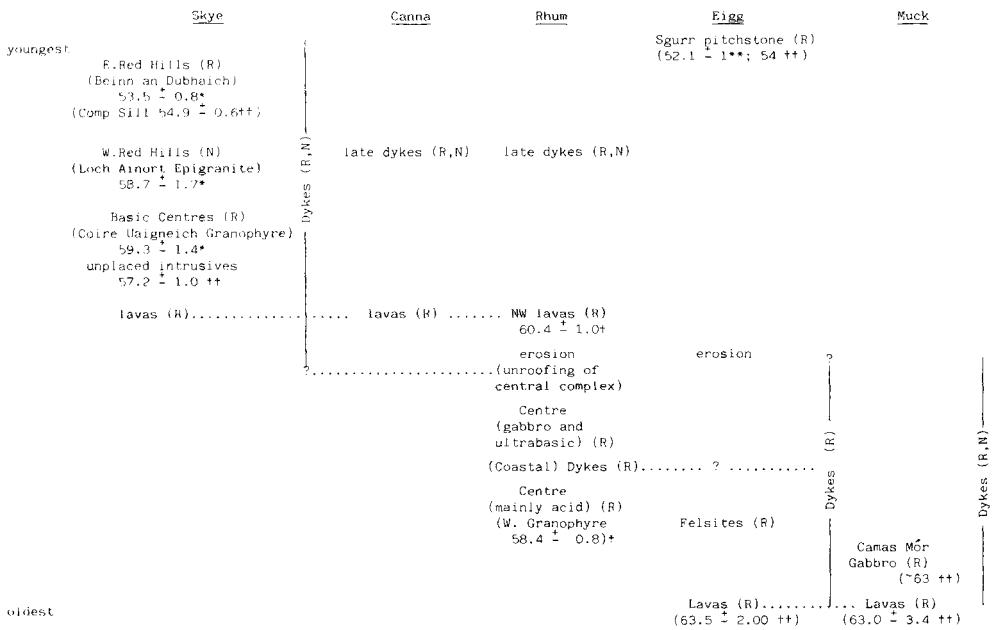
The palaeomagnetic results for lavas, dykes and intrusive bodies from the islands of Muck and Eigg, which had not been sampled previously for palaeomagnetic measurement, are presented and then these results are combined with all the available evidence to show the relationship between the Small Isles and Skye.

Geology

On both Muck and Eigg the exposed rocks are almost entirely Tertiary plateau lavas (Harker 1908). The base of the lava piles can be seen to rest on pre-Tertiary rocks, i.e. Jurassic sediments on the shore of Camas Mór in Muck and Cretaceous sediments below the basal and Laig lava groups in Eigg. The structure and petrology of the lavas have been examined by Allwright (1980); the majority (~ 80 per cent) of them are basaltic and the remainder are differentiates (hawaiites and mugearites) (Allwright 1980 and private communication). Chemically the lavas of Eigg and Muck are broadly similar to those of the Skye Main Lava Series defined by Thompson, Esson & Dunham (1972) and Mull sequences, though a few are similar to the Preshal Mor and Fairy Bridge magma types of Skye (Allwright 1980). Within the Muck sequence several bands of red material, which are seen to overlie weathered basalt and infill cracks in the upper surface of the underlying flow, provide useful marker horizons. The origin of these 'red beds' is uncertain but they may be altered tuffs originating from an explosive event (Allwright, private communication). In Eigg the most conspicuous horizon is a distinctively coloured flow, the Grey Rock, classified as a mugearite by Allwright (1980).

The lava piles are cut by dyke swarms which, typically of the BTIP, trend approximately NW–SE. The dykes have been mapped by Speight (1972) who showed that the axis of the intense swarm cutting Muck runs from Ardnamurchan to SE Rhum. The Eigg dyke swarm is

Table 1. The Tertiary geological sequence and magnetostratigraphy for the Small Isles and Skye.



Based on information from Allwright (1980), Dagley & Mussett (1980, 1981), Dickin (1981)*, Dickin & Jones (1983)**, Emeleus & Forster (1979), Meighan *et al.* (1982), Mussett (1984)†, Wilson *et al.* (1972b, 1974, 1982) and this laboratory††.

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numerically less intense than that of Muck; it appears to die out to the SE toward the mainland but to the NW it intersects SE Rhum. It is possible that the dykes are related to the Rhum Centre but as some are seen to be truncated by gabbro bodies in Rhum (Emeleus 1973 and Emeleus, private communication) they were intruded after the acid but before the basic parts of that centre. Some are seen to cut the Grey Rock of Eigg but others do not, so dyke intrusion and lava extrusion proceeded in parallel and presumably at least some of the dykes were feeders to lavas. Thus the dykes and lavas of Eigg and Muck must be at least as old as the Rhum Centre, that is $\gtrsim 58$ Myr old.

On Muck the only other intrusive body exposed is the Camas Mór gabbro, a dyke-like body (~ 70 m thick) which also trends NW–SE but as this body is cut by dykes of the regional swarm it must predate at least some of the swarm. The Eigg lavas are intruded by three felsite bodies, the Sgorr Sgaileach (northern), the Sronn Laimhige (Laig) and the Grulin felsite, all of which preceded most but not all of the Eigg dykes.

The youngest igneous body on Eigg is the Sgurr pitchstone (Bailey 1914). This is a lava emplaced over a conglomerate in channels eroded in the main lava pile (Allwright & Hudson 1982). A radiometric age of ~ 52 Ma (Dickin & Jones 1983) suggests that this is probably the youngest igneous body on the Small Isles.

The relative geological sequences in the Small Isles are shown in Table 1, which is based on all the available information.

Palaeomagnetic sampling and results

Sampling locations are shown in Fig. 1. At each site four or six 25 mm diameter cores were drilled and oriented relative to the Sun or a geographic marker. Specimens about 25 mm long cut from the cores were measured using a parastatic magnetometer and 'cleaned' using alternating field (af) and/or thermal demagnetization. Af demagnetization was routinely done to 40 mT in 5 mT steps and extended to higher peak fields up to 1400 mT when necessary. Thermal demagnetization was carried out in 50°C steps to the Curie temperature. Thermal demagnetization appears to be less efficient than af demagnetization in that sometimes the changes in direction from the NRM were less and agreement between the end points for individual cores was poorer. However, in general, the two processes give similar results and only the af results are used here. In almost all cases, stable directions which clustered well were reached after a small number of demagnetizing steps. There was no evidence for secondary components. Site mean directions were calculated using the method of Dagley & Ade-Hall (1970). Small pieces from each core were heated rapidly to about 700°C to determine the Curie point. The chief features of the directional and thermomagnetic results which are given in Table 2 and Figs 2 and 3 are discussed below.

MUCK LAVAS

The sequence of lavas on Muck (Table 3a) was sampled from the shore near Sgorr an Laogh (40407877) to the summit of Beinn Airein (40307915) (Fig. 1a), augmented by samples from other sections mapped by Allwright (1980). The most useful horizons for correlating sequences in Muck are the 'red beds'. Our composite sequence may not include the oldest flows but the base is estimated to be no more than 20 m below site 764 (flow 10′ of Allwright). Allwright's mapping showed that it is not possible to match all the individual flows between time-equivalent lava sequences even though these are separated by as little as half a kilometre. There are also differences between directions of magnetization and thermomagnetic behaviour for time-equivalent sequences. This evidence is consistent with

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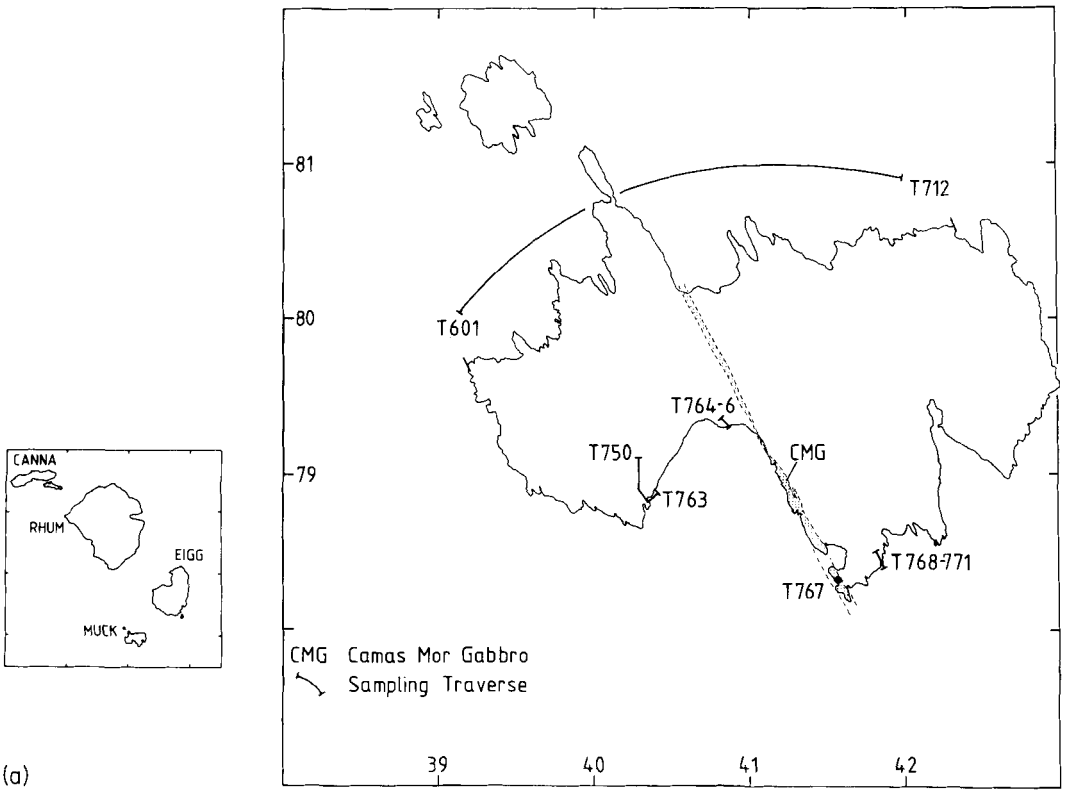


Figure 1. Outline maps showing location of sampling traverses. (a) Muck, (b) Eigg.

Allwright's suggestion that extrusion occurred spasmodically from several different local sources. However, in view of the different results obtained for a single unit (viz. the Grey Rock lava of Eigg) directional matching must be treated with caution.

All 21 flows or flow units sampled are reversely magnetized and the unit mean direction of magnetization (Fig. 3a) for each has α_{95} less than 7.0° (Fig. 2a). The group mean direction is shallower than a centred axial dipole field (Table 2, Fig. 4), with a far-sided pole.

The majority of samples show evidence of two Curie points when they are first heated (Table 2). Of the remainder some show a low value and some a high value. On cooling, the Curie points for those with low value are higher but still significantly different from the high values. This behaviour suggests patchy intermediate deuteric oxidation and only moderate regional hydrothermal alteration implying only light burial (Ade-Hall *et al.* 1971). Together with the absence of secondary components of magnetization this indicates that the directions of magnetization are primary. There do not appear to be any correlations of Curie point with rock type or stratigraphic position.

MUCK DYKES

A representative collection across the swarm was obtained by sampling the dykes exposed along the northern coast. Speight (1972) mapped 113 dykes between Rubh' Lean Na Laraich (391797) and Port nam Maol (424816) (Fig. 1a) and we set out to sample alternate dykes. We have followed Speight's numbering (with the addition of 600 to distinguish them

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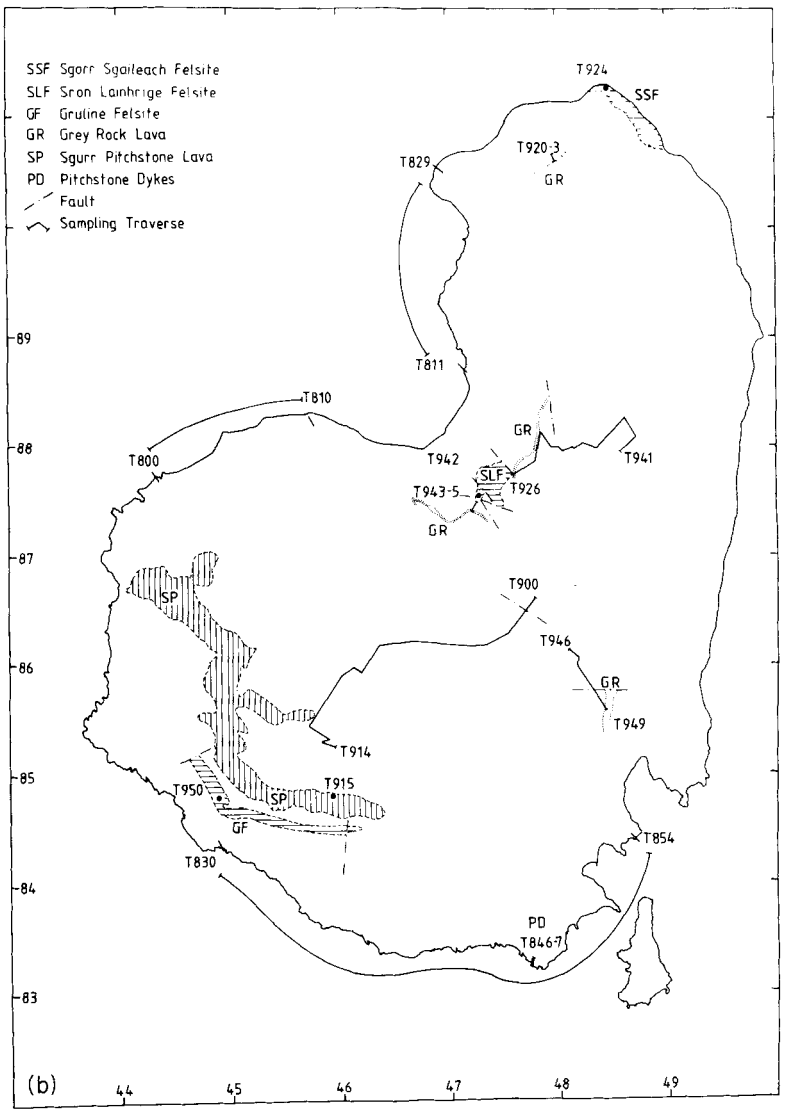


Figure 1 – *continued*

from other dykes of the Small Isles) though in a few cases individual dykes sampled were not certainly identified with Speight's.

Of the 52 dykes sampled, four are normally magnetized, 36 reversely magnetized and 12 intermediate (Table 2, Fig. 3b) on the basis of the classification of Wilson, Dagley & McCormack (1972a) (polar colatitute $\theta < 40^\circ$ normal, $40^\circ < \theta < 140^\circ$ intermediate, $\theta > 140^\circ$ reversed). The intermediate directions show no recognizable bias towards positive or negative inclinations, but the reversed directions are biased to shallow inclinations and poles scatter around the locus expected for a centred axial dipole source. The directions of magnetization of the normally magnetized dykes are not close and there does not appear to be any correlation of polarity with location.

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Table 2. (a) Palaeomagnetic results.

Code	Samples	D (°ETN)	I (°down)	α_{95}	Polarity	T_c (°C)	
geocentric axial dipole		0	±72				
Muck lavas							
A03T750	4	173.9	-55.9	3.4	R	360	470
751	4	172.6	-57.1	2.2	R	395	460
752	4	177.7	-48.2	4.8	R	410(500)	525
753	4	180.3	-65.9	1.1	R	380(500)	475
754	4	187.0	-73.6	1.9	R	395(545)	525
755	4	171.7	-66.8	0.9	R	590	565
756	4	192.9	-68.6	2.6	R	390	515
757	3	192.4	-64.6	1.0	R	560	540
758	4	185.1	-70.7	3.1	R	430	510
759	4	166.8	-62.1	2.8	R	370	520
760	3(4)	165.8	-59.2	1.4	R	560	600
761	3	185.5	-56.2	5.9	R	570	550
762	4	186.5	-60.5	2.4	R	420(560)	520
763	4	191.6	-57.5	2.1	R	360	460
764	4	199.2	-67.7	2.9	R	375	490
765	4	158.5	-63.7	3.8	R	430(560)	550
766	4(5)	158.5	-63.5	6.1	R	390(560)	535
768	4	178.3	-70.6	1.5	R	580	580
769	4	150.0	-64.6	2.4	R	435(565)	520
770	3	166.7	-63.1	1.3	R	420(530)	540
771	3	167.5	-69.1	2.9	R	428(550)	540
group mean 21 units		176.3	-63.8	3.3	R	——	
(pole; lat. 78.4, long. 186.9, dp = 4.2, dm = 5.2)							
Muck dykes							
A03T601	4	350.9	67.2	2.4	N	325	——
603	4	164.0	-45.9	5.2	R	470	550
605	4	200.8	-51.8	4.2	R	410	515
607	4	189.2	-74.6	0.7	R	335	500
609	4	192.9	-37.7	6.2	R	485	530
611	3(4)	177.3	-71.5	13.4	R	480	500
613	4	198.7	-52.8	2.7	R	350	500
615	4	206.6	-63.3	2.5	R	320	520
617	4	269.0	-22.9	2.3	I	410	450
627	2(4)	176.8	-12.7	4.3	I	405	495
636	4	182.6	-52.3	3.0	R	460	510
638	4	199.3	-48.8	1.8	R	390	500
639	4	203.0	-71.7	8.6	R	480	505
641	4	350.9	62.4	2.3	N	555	540
643	4	215.6	-24.9	5.9	I	410	500
645	4	9.9	72.0	1.5	N	350	490
647	3(4)	144.6	-52.1	3.9	R	590	590
649	4	168.2	-35.5	2.8	R	480	480

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Table 2. (a) – *continued*

Code	Samples	D (°ETN)	I (°down)	α_{95}	Polarity	T_c (°C)	
Muck dykes continued...							
A03T651	4	177.1	-35.9	2.7	R	470	—
653	4	172.4	-38.7	4.4	R	570	570
654	4	163.5	-57.7	2.0	R	400	—
656	4	195.8	-56.4	0.5	R	450	—
657	4	172.2	-52.1	4.8	R	380	450
659	4	216.5	-48.1	2.9	R	370	460
661	4	177.8	-64.8	0.7	R	360	—
662	4	179.0	-40.7	3.1	R	330	—
663	4	137.7	-32.5	5.3	I	360	520
667	2	203.2	-35.4	2.0	I	400	505
669	4	119.2	-61.8	9.2	R	330	510
671	4	174.9	-48.2	2.0	R	400	530
673	3(5)	204.7	-62.7	4.5	R	430	500
675	4	206.4	-69.6	1.5	R	540	540
677	4	159.9	-64.2	2.2	R	530	530
679	4	181.5	-40.1	2.6	R	400	520
681	4	162.4	-46.9	3.8	R	360	500
683	4	185.1	33.3	2.7	I	400(520)	525
685	3(4)	220.8	-42.0	10.3	I	400	520
687	4	169.6	48.2	17.4	I	440	500
689	4	194.2	-58.6	1.4	R	380	500
691	4	203.8	-59.8	7.7	R	520	520
693	4	90.7	65.2	4.8	I	410	510
695	4	2.9	77.4	3.0	N	440	340
697	4	201.0	5.7	4.5	I	295	—
699	4	206.2	-3.1	4.2	I	250	—
700	4	204.8	-30.3	8.4	I	390	510
702	4	157.0	-47.8	3.7	R	510	510
704	4	253.4	-47.5	6.6	I	450	520
705	4	191.4	-32.3	1.6	I	510	510
706	4	207.9	-8.5	8.1	I	380	510
708	4	227.3	-40.7	5.5	I	470	515
710	4	195.8	-29.6	2.1	I	405	490
712	4	189.5	-35.4	4.4	R	480	500
group means, $\alpha < 10°$							
N	4 units	357.0	69.9	8.2	N		
	(pole; lat. 86.4, long. 203.4, dp 12.1, dm 14.1)						
I	16(17) units						
R	30(31) units	181.2	-54.2	5.4	R		
	(pole; lat. -67.8, long. 351.2, dp 5.3, dm 7.6)						
Camas Mór gabbro dyke							
A03T767	3(5)	165.2	-32.3	1.9	I	520	535

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Table 2. (a) – *continued*

Code	Samples	D (°ETN)	I (°down)	α_{95}	Polarity	T_c (°C)	
Eigg lavas							
A03T900	4	161.7	-56.1	2.5	R	480	495
901	4(6)	216.3	-57.2	7.9	R	485	475
902	5(6)	197.8	-35.6	4.2	R	480	480
903	3	190.0	-42.5	1.0	R	545	505
904	4	210.1	-60.2	5.8	R	350(530)	495
905	4	186.5	-56.7	1.3	R	505	500
906	4	164.8	-44.0	3.4	R	485	475
907	4(5)	180.4	-72.1	1.3	R	355(500)	495
908	4	187.2	-53.9	0.8	R	380(480)	475
909	3(4)	217.2	-50.6	10.9	R	480	480
910	3(4)	184.4	-43.1	4.2	R	345(495)	505
911	0(4)	disparate directions, no mean			R?	565	535
912	3(6)	165.5	-65.5	1.6	R	475	460
913	4	160.4	-73.2	3.1	R	555	535
914	4	187.7	-51.4	0.8	R	350(550)	475
920	4	194.8	-61.8	2.0	R	400	510
921	4	138.8	-62.5	0.8	R	555	545
922	4	140.3	-60.2	3.2	R	330	555
923•	4	178.0	-60.9	1.8	R	585	—
926	4(5)	167.0	-52.3	6.8	R	480	—
927*	4	175.9	-42.6	1.4	R	560	525
928	4	209.2	-71.3	3.5	R	370(460)	520
929	4	150.7	-24.1	2.5	I	260(500)	500
930	4	195.7	-52.1	1.7	R	505	505
931	4	196.6	-48.3	2.0	R	370(500)	460
932	4	160.2	-59.5	4.6	R	345	500
933	4	185.7	-50.8	1.1	R	375(485)	490
934	4(5)	158.1	-62.9	1.4	R	570	550
935	4	154.7	-60.9	1.2	R	560	510
936	2(4)	167.1	-58.5	0.4	R	360(565)	525
937	4	164.1	-60.8	0.6	R	250(550)	520
938	4(6)	189.4	-48.0	11.1	R	355(245)	510
940	4	165.8	-63.6	0.4	R	380(530)	515
941	4	203.4	-76.9	2.4	R	355(520)	490
943	4	199.5	-62.1	1.6	R	540	530
944	4	169.6	-58.2	2.0	R	550	550
945*	6	173.7	-71.5	13.1	R	540	545
946	4	209.6	-77.0	0.9	R	375(545)	520
947	4	184.4	-54.2	2.6	R	375(545)	500
948	4(6)	187.1	-29.9	5.6	I	?	?
949*	5	177.7	-26.9	3.4	I	355(385)	500

• Grey Rock

group means, $\alpha < 10°$

R 34 units 179.7 -59.3 4.3 R
(pole; lat. -73.2, long. 354.5, dp 4.8, dm 6.4)

I 3 units 171.6 -27.9 26.5 I
(pole; lat. -47.4, long. 5.9, dp 15.9, dm 29.0)

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Table 2. (a) – *continued*

Code	Samples	D (°ETN)	I (°down)	α_{95}	Polarity	T_c (°C)	
Felsite intrusions							
Sgorr Sgaileach							
A03T924	4(6)	254.0	-81.6	10.4	R	——	——
Sronn Laimhige							
942	3(6)	238.4	-52.3	1.7	I	——	550
Grulin							
950	5(6)	164.3	-48.7	2.8	R	510	510
Sill (cutting 924)							
925	4	194.8	-76.5	0.5	R	350	500
Eigg dykes							
A03T800	4	166.3	-67.3	2.3	R	320	540
801	4	153.2	-57.3	2.4	R	400(570)	570
802	4	165.0	-62.9	3.1	R	265(330)	500
803	4	169.9	-63.0	2.8	R	335(430)	530
804	4	173.3	-59.0	3.5	R	? (520)	520
805	4	163.9	-58.8	2.1	R	300	515
806	5	169.0	-60.2	1.6	R	300(500?)	500
807	4	140.6	34.1	11.6	I	360(410)	510
808	4	186.7	-36.0	3.9	R	320	485
809	3(4)	179.9	-42.5	0.8	R	360(550)	530
810	2	165.8	-23.3	3.0	I	390	525
811	4	197.5	-74.4	1.0	R	330	525
812	4	197.2	-68.6	1.0	R	355	——
813	4	223.1	-81.7	1.8	R	430	?
814	4	195.0	-59.1	3.0	R	440(575)	530
815	3	157.7	-63.5	15.3	R	470	540
816	4	184.8	-49.7	24.0	R	350(575)	?
817	3	178.0	-64.5	14.5	R	450	535
818	4	180.3	-29.8	10.6	I	400(500)	520
819	1	only one core, no mean			–	?	580
820	4	235.4	-70.7	0.8	R	?	540
821	4	194.5	-69.7	2.3	R	500	540
823	4	180.5	-60.3	4.4	R	380(500)	530
824	1	only one core, no mean			R	——	——
825	4	174.2	-60.6	3.6	R	460	550
826	4	177.2	-69.0	4.2	R	600	600
827	4	169.3	-86.1	6.6	R	450(600)	550
828	4	171.9	-66.5	2.0	R	470	?
829	4	157.0	-61.2	6.1	R	330(600)	600
830	4	142.6	-25.9	5.6	I	——	——
831	3(4)	182.7	-64.7	2.7	R	425	500
832	4	175.6	-54.9	7.2	R		580
833	4	202.4	-71.3	2.9	R	450	510
834	4	170.5	-59.5	3.0	R	380	500
835	5	167.1	-45.5	2.8	R	350(550)	550
836	4	185.3	-63.4	1.7	R	290	510
837	4	220.0	-54.8	1.6	R	350	485

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Table 2. (a) – *continued*

Code	Samples	D (°ETN)	I (°down)	α_{95}	Polarity	T_c (°C)	
Eigg dykes continued...							
A03T838	4	170.4	-65.0	3.3	R	420	540
839	4	145.7	-55.5	6.0	R	370	490
840	4	191.8	-54.4	4.6	R	430	540
841	4	192.8	-33.0	1.9	I	430	540
842	4	234.7	-65.9	2.5	R	420	480
843	4	191.4	-56.9	3.9	R	470	530
844	1	only one core, no mean			R	—	—
845	4	183.5	-38.6	1.7	R	400?	500
846	4	205.4	-40.7	3.6	R	350?	550?
847	4	216.1	-45.5	8.2	R	—	—
848	4	187.7	-56.0	1.2	R	380	520
849	4	218.5	-72.9	1.9	R	335	500
850	5	disparate directions, no mean			R?	425	505
851	4	196.1	-50.0	2.6	R	520	530
852	4	186.0	-34.6	2.9	R	430	570?
853	4	221.3	-57.4	3.6	R	420	545
854	4	185.9	-51.8	4.4	R	275	515
group means $\alpha < 10°$							
R	42 units	185.5	-60.9	4.2	R		
(pole; lat. -74.9, long. 338.2, dp 4.9, dm 6.4)							
I	3 units	166.4	-28.9	35.5	I		
(pole; lat. -47.3, long. 13.3, dp 21.5, dm 39.1)							
Sgurr pitchstone lava							
A03T915	3(6)	171.1	-69.6	3.2	R	360	500

Key: Samples – number used to calculate unit mean (number measured, if different), one sample from each core.

Polarity { Normal, Intermediate, Reversed } polar colatitude $0° < \theta < 40°$, $40° < \theta < 140°$, $140° < \theta < 180°$ (Wilson *et al.* 1972a)

T_c strong field Curie points on heating and on cooling. Second (high) Curie point in brackets.
σ standard deviation.
ϵ standard error on mean.
n number of values.

The mean direction of magnetization for the reversely magnetized samples is significantly shallower than the mean direction for the lavas, for which the individual mean directions are more tightly grouped (Fig. 4).

Thermomagnetic treatment shows that in most cases the behaviour is simple with a single Curie point but there is a large spread of values from 250° to 570°. On cooling, samples with low Curie points on heating (< 500°) show the onset of magnetization to occur at ~ 500°.

CAMAS MÓR GABBRO

The Camas Mór Gabbro was sampled near its southernmost exposure. Its mean direction of magnetization classifies it as intermediate (polar colatitude 41.8°) though the circle of confidence includes the class boundary.

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Table 2. (b) Group mean Curie temperatures.

			low				high			
			T_c	σ	ε	n	T_c	σ	ε	n
Muck lavas	single low	h	376	26	6	19				
		c	467	59	13	19				
	single high	h					581	11	5	5
		c					557	21	9	8
	double	h	416	46	6	55	552	23	4	46
		c					525	34	5	41
Muck dykes	single low	h	423	78	7	110				
		c	510	36	4	98				
Eigg lavas	single low	h					520	39	3	46
	double	h	355	28	4	43	519	32	6	34
	all	c					583	30	3	91
Eigg dykes	single low	h	368	61	9	46				
	single high	h					498	49	12	17
	double	h	274	58	10	34	512	72	15	22
	all	c					525	34	2	102

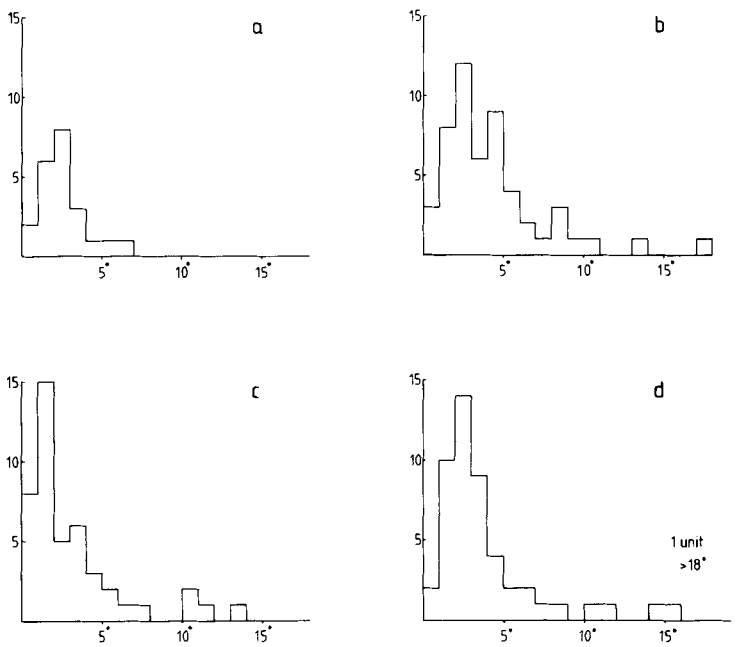


Figure 2. Histograms of α_{95}. (a) Muck lavas, (b) Muck dykes, (c) Eigg lavas, (d) Eigg dykes.

EIGG LAVAS

On Eigg two main and two subsidiary lava sections were sampled (Fig. 1b and Table 3b) using the mapping of Allwright (1980). The lowest flow sampled (920) is very close to the base, if not actually the first extrusion.

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Directions of magnetization for individual cores from two of the units were too scattered to yield mean directions and all but one of the remainder are reversely magnetized, the exception having a shallow negative dip and so is classed as intermediate (Table 2, Fig. 3c). The lava pile is estimated to tilt about 6°E (Allwright 1980 estimates 5°–10° to the south). If this tilt followed magnetization then (by restoring the lavas to the horizontal) the original direction of magnetization was even shallower with respect to a centred axial dipole field.

Only one unit shows a simple thermomagnetic curve with a single low Curie point (< 400°C). The rest divide equally between those with 'double' Curie points and those with a single high value. The average values on heating and on cooling show that all the high Curie points could belong to the same population and it may be that all are intrinsically double but in some cases the lower one has not been resolved.

In each of the two principal sequences sampled, the unit mean directions of each flow are distinct from those above or below. Where, in the western (Post Office) sequence, there are flows with similar directions they are demonstrably different flows by position, petrological characteristics and thermomagnetic behaviour. In the northern (Cleadale) sequence four suc-

Table 3. (a) Muck lava sequences (sequences from Allwright (1980) with palaeomagnetic sample number T750-771) (see Fig. 1 for locations).

Beinn Aireinn			Sgorr nan Laogh		Torr Nam Firheach		Port an T-Seilich	
	unit	sample no.	unit	sample no.	unit	sample no.	unit	sample no.
		750						
1	M630	751						
red bed	HM 7563							
1a	HM 7564	752						
2	M 624	753						
3	M 629	754						
	HM 7571							
red bed	M 631							
	HM 7572							
4	M 622	755						
red bed	M 623							
5	M 627	756						
	HM 7574							
red bed	-							
6	M 620	757	HM 7536		HM 7548			
	HM 7576				HM 7547	(6.5)		
red bed	-		HM 7535					
							HM 75142	(6.8) 771
							M 657	(6.9) 770
7	HM 626	758	-		HM 7546			
	HM 7577						HM 75143	769
8	M 618	759	HM 7534	-	HM 7544	766	HM 75149	
	HM 7578							
	-	760					M 656	(8.5)
9	-	761	HM 7533	-				
					HM 7543	(9.5) 765	HM 75148	(9.5')
10			HM 7631	762	HM 75127	(10') 764	M 653	(10') 768
					M 605		HM 75146	
11			HM 7538	763			HM 75144	(11')
					sediment			

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Table 3. (b) Eigg lava sequences (sequences from Allwright (1980) with palaeomagnetic sample number T900-949) (see Fig. 1 for locations).

North Cliffs	Cleadale–Beinn Bhuidhe (northern)	Laig Gorge	Post Office–Cora Bhein (western)
			E 668 – 910
			E 665 – 911
			E 664 – 912
			E 667 – 913
			E 663 – 914
			E 641 – 909
	top of Beinn Bhuidhe		E 639
	HE 7493		E 638 } – { 908
	HE 7492		E 636 } – { 907
	HE 7491 – 941		E 635 } { 906
	HE 7490 – 940		E 634
	HE 7489 – 938		E 632
	HE 7488 – 937		E 608 } { 905
	HE 7487 – 936		E 607 } { 904
	HE 7486 – 935		E 606 } – { 903
	HE 7485 – 934		E 605 } { 902
	933		{ 901
HE 7429	HE 7483 – 932		E 604
HE 7428	HE 7482 – 931		E 601 – 900
–	HE 7481 } { 930		E 642 – 946
HE 7427	HE 7419 } { 929		E 645 – 947
HE 7426	HE 7480 } – {		E 644 – 948
HE 7425	HE 7479 } { 928		fine basalt –
Grey Rock HE 7421 – 923	= HE 7478 – 927	= Grey Rock – 945	= Grey Rock 949
HE 7430 } { 922	HE 7416 – 926	basalt – 944	
HE 7431 } – { 921		columns – 943	
HE 7432 } {			
HE 7422 – 920			
sediment HE 7495			

cessive units (934, 935, 936, 937) do form quite a close directional group and they are also of a similar petrological type (Allwright 1980). Although these two sequences should span a similar time period there is no obvious similarity between the two. So, unlike Skye (Khan 1960; Wilson *et al.* 1972b) and northern Ireland (Wilson 1970), no lateral palaeomagnetic correlations can be made. However, possible correlations are 908 with 938 which have similar petrology and Curie point behaviour. The two subsidiary sections at the base of the sequence have general features in common but again there is no direct match.

The Grey Rock lava was sampled at four different sites, 923, 927, 945 and 949, which gave different directions of magnetization. Although the large uncertainty on the site-mean direction of magnetization of 945 means that it cannot be distinguished from 923, neither of these two units have directions which match either of the other two sites for this unit, 927 and 949 which themselves disagree. No evidence was found that these sites had been affected by lightning but the possibility cannot be ruled out. Faulting or remagnetization by intruding bodies are alternative explanations but no supporting evidence was found. This variation of direction within a single unit, because of its importance, is the subject of a separate investigation involving a Mull Plateau Lava (Dagley, Mussett & Skelhorn 1984a) and we will restrict ourselves to polarity matching in this paper.

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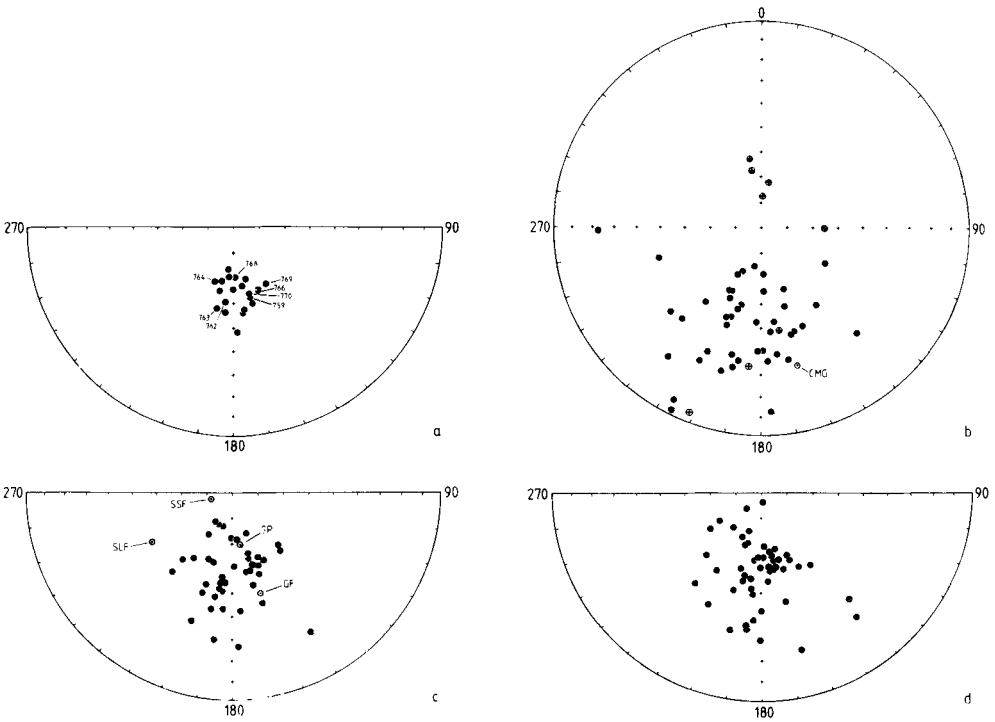


Figure 3. Equal area stereographic projections of unit mean directions of magnetization: negative inclinations ● ⊗, positive inclinations ⊕. (a) Muck lavas, (b) Muck dykes, (c) Eigg lavas and felsites, (d) Eigg dykes.

THE FELSITES

All three felsites, and a sill (925, an aphyric basalt) cutting the northernmost, were sampled. The three felsite bodies yield different mean directions of magnetization (Fig. 3c); one is intermediate with negative dip, the other two reversed. However, the Sgorr Sgaileach (924) and Sronn Laimhige (942) felsites are weakly magnetized, the former having a poorly defined mean direction and it is possible that the primary directions have not been revealed. The sill gives a precisely defined direction not far from the poorly defined direction of the felsite it intrudes and the latter may have been remagnetized by the sill.

Only the Grulin felsite (950) has a clearly defined Curie point (510°C).

THE EIGG DYKES

Because of poor accessibility the Eigg dyke swarm could not be sampled across a single section so, to give a representative collection, roughly every other dyke mapped by Speight (1972) was sampled between Rubha an Fhasaidh (439874) and Bogha na Brice-Nis (465904) on the NW facing coast and from near Sgeir Eskernish (449843) to Na Gurra Bain (487844) along the south coast (fig. 1). There should be little, if any, duplication in these sections because of the NW–SE trend of the swarm.

No normally magnetized dykes were found and of the 54 sampled, 48 are reversely magnetized, five intermediate. The remaining one gave directions of magnetization for individual cores which were too scattered to average but which indicate reversed polarity

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(Table 2, Fig. 3d). The distribution of unit mean directions is asymmetric; those falling to the west of south are more dispersed to both high and low inclinations, while those to the east are more closely clustered (about $-65°$). This distribution gives equivalent poles which are right-handed and far-sided for about half the units, farsided but left-handed for a little less than half and near-sided poles for a few.

On heating samples from the dykes some exhibited two Curie points but many were single and some vague and ill-defined. The mean values on heating and on cooling suggest that the single low and double low values could be from the same population, distinct from the single high values. The two pitchstone dykes, 846 and 847, have heating curves with very gradual slopes so that their Curie points are poorly defined. They appear, however, to be low, ~ 300–350°C on heating and 500–600°C on cooling.

THE SGURR PITCHSTONE

This is reversely magnetized with a direction of magnetization 171.1°, −69.6°, $\alpha_{95} = 3.21°$ and the thermomagnetic curves show two Curie points at ~ 360° and 500°C. These results distinguish this body from the two pitchstone dykes sampled and from the Grulin Felsite which is consistent with Allwright's (1980) conclusions and also support the isotopic evidence (Dickin & Jones 1983) that the Grulin Felsite was not a feeder for the Sgurr.

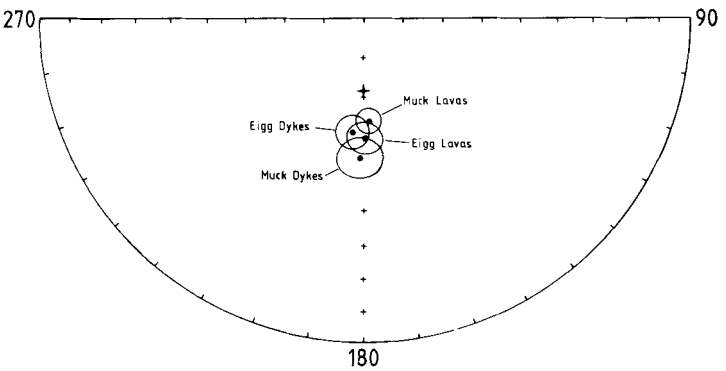


Figure 4. Equal area stereographic projection of mean directions of magnetization for reversely magnetized groups. Direction of centred axial dipole field +.

Radiometric dates

The only previously published, reliable date is for the Sgurr of Eigg, by the Rb-Sr isochron method which gave an age of 52.1 ± 1.0 (2σ) Ma (Dickin & Jones 1983).

Several rocks from Eigg and Muck have been analysed by this laboratory using the $^{40}Ar-^{39}Ar$ step-heating method. Details are as given in Mussett, Ross & Gibson (1980b). Results are given in Table 4 and Fig. 5. A sample of the Sgurr gave an age spectrum that is slightly convex, apart from the first two, very small, steps. No combination of successive steps fully satisfies the criteria of Lanphere & Dalrymple (1978) (see Table 4b), but most of the step ages are about 53 Myr in agreement with the date of Dickin & Jones.

A sample of T923, a lava near the base of the lava succession, at the north of Eigg, has an age spectrum that, on the whole descends, again apart from the first two small steps. However, steps 5–9 satisfy the criteria and yield an age of 63.5 ± 2.0 Myr. Thus igneous activity

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in Eigg spanned about 10 Myr, the whole duration of activity in the British Tertiary Igneous Province (see Table 1).

A sample of one of the basal lavas of Muck, T765, gave a spectrum that first rises, then falls, before becoming level. Steps 7–12 fulfil the acceptance criteria, though the $^{40}Ar/^{36}Ar$ contamination ratio differs from the atmospheric ratio at the 1σ level, though not the 2σ

Table 4. (a) $^{40}Ar-^{39}Ar$ dating. (a) Step-heating analyses.

Step	Age (±2σ) Ma	$\%^{40}Ar$ (rad)	$\%^{39}Ar$ released	$^{40}Ar/^{39}Ar^*$	$^{36}Ar/^{39}Ar^*$ units of 10^{-3}	$^{37}Ar/^{39}Ar^*$
Eigg: Sgurr (irradiation no. R1486)						
1	386 ± 13	66	0.9	321	371	no ^{37}Ar detectable
2	420 ± 6	73	0.7	321	297	"
3	51.0 ± 0.7	79	15.0	32.6	23.7	"
4	53.8 ± 0.9	84	11.5	32.2	17.7	"
5	55.4 ± 0.9	86	4.7	32.3	15.1	"
6	54.2 ± 0.8	85	12.8	32.2	16.7	"
7	54.8 ± 0.8	85	7.6	32.6	17.1	"
8	55.4 ± 1.0	73	11.4	38.0	34.4	"
9	56.3 ± 0.8	87	5.3	32.7	14.7	"
10	55.1 ± 0.9	85	7.0	32.6	16.7	"
11	53.3 ± 0.7	82	8.3	32.6	19.9	"
12	53.4 ± 0.9	82	14.9	32.8	20.2	"
Eigg: basal lava (irradiation no. R1598)						
1	-23 ± 9	-1	0.3	378	1297	1.03
2	55.0 ± 2.1	24	0.7	53.1	137	0.445
3	69.7 ± 2.3	68	1.8	23.8	35.8	0.293
4	64.5 ± 2.1	89	8.8	16.8	6.11	0.211
5	63.7 ± 2.0	95	16.4	15.6	2.84	0.158
6	63.1 ± 2.0	96	10.7	15.3	2.14	0.114
7	63.5 ± 2.0	95	3.7	15.6	2.83	0.005
8	63.4 ± 2.0	91	20.8	16.2	5.16	0.175
9	63.5 ± 2.0	91	10.8	16.1	4.68	0.278
10	62.5 ± 2.0	94	1.7	15.4	3.09	0.309
11	63.1 ± 2.0	92	4.2	16.0	4.57	0.293
12	62.4 ± 2.0	87	13.3	16.7	7.62	0.564
13	62.4 ± 2.0	81	6.4	17.9	11.7	0.621
14	-10 ± 10	-1	0.4	387	1318	0.795

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Table 4. (a) – *continued*

Step	Age (± 2) Ma	$\%^{40}Ar$ (rad)	$\%^{39}Ar$ released	$^{40}Ar/^{39}Ar^*$	$^{36}Ar/^{39}Ar^*$ units of 10^{-3}	$^{37}Ar/^{39}Ar$
Muck: basal lava (irradiation no. R.1600)						
1	52 ± 17	3	0.2	482	1590	–
2	91.0 ± 3.8	16	1.1	142	405	5.60
3	103.3 ± 3.3	31	1.3	83.3	197	5.34
4	105.2 ± 3.3	35	1.1	74.7	166	5.87
5	79.0 ± 2.1	60	7.4	32.2	43.8	2.73
6	67.2 ± 1.8	66	7.8	25.2	29.6	1.56
7	59.5 ± 2.6	40	8.8	36.2	73.8	2.52
8	62.5 ± 1.6	78	23.0	19.7	15.4	2.86
9	61.1 ± 2.5	59	0.7	25.5	35.4	–
10	63.2 ± 1.6	82	39.2	18.8	12.3	4.81
11	64.9 ± 2.0	80	4.0	19.9	14.9	5.68
12	62.4 ± 1.6	73	5.4	21.0	20.6	5.65
Muck: Camas Mor Gabbro (irradiation no. R.1597)						
1	266 ± 10	41	1.3	157	312	2.49
2	158 ± 6	53	1.6	70.0	112	2.74
3	84 ± 3	65	4.6	29.9	36.2	3.11
4	70.7 ± 2.7	41	5.3	39.9	81.4	5.11
5	63.4 ± 2.4	75	17.9	19.5	18.3	6.56
6	63.2 ± 2.4	80	13.3	18.1	13.9	6.30
7	63.1 ± 2.6	86	6.4	16.8	9.25	4.88
8	22.9 ± 2.3	85	7.9	17.0	9.72	3.96
9	61.1 ± 2.2	88	4.7	16.0	7.45	3.18
10	60.1 ± 2.3	88	4.0	15.7	6.60	0.372
11	59.9 ± 2.3	88	2.6	15.6	6.74	2.17
12	61.5 ± 2.2	88	6.0	16.0	6.75	2.01
13	69.1 ± 2.5	82	24.4	19.3	13.3	6.37

* Corrected for mass discrimination and, for the $^{37}Ar/^{39}Ar$ ratio, for decay of ^{37}Ar.

level. The plateau age of 63.0 ± 3.4 Ma is believed to be a good estimate of the age of extrusion of the lava. This value is concordant with that for the basal lava of Eigg suggesting they are contemporaneous, and with the tentative date for the Camas Mór dyke, given in the next paragraph.

The spectrum of the great Camas Mor gabbro dyke is erratic and the changing value of the $^{37}Ar/^{39}Ar$ ratio suggests that two or more minerals are releasing argon, over a different range of temperatures. Though no combination of successive steps satisfy the criteria the mean age of the short plateau formed by steps 5–8 is 63.1 ± 2.3 Ma; this suggests that this dyke is not significantly different in age from that of the lavas.

These dates are consistent with the known stratigraphic relations and help to quantify them (Table 1).

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Table 4. (b) Dating summary data.

Sample	Irradiation No.	Total fusion age, Ma	Steps	% ^{39}Ar released	Plateau age ($\pm 2\sigma$), Ma	Correlation plot: Age ($\pm 2\sigma$), Ma	Correlation plot: $^{40}Ar/^{36}Ar$ contaminant ratio ($\pm 2\sigma$)	MSWD	Remark
Eigg, Sgurr	R 1486	58.8	4 - 10	60	53.8 ± 0.9	55.3 ± 1.1	264 ± 18	8.8	not acceptable
Eigg, basal lava T923	R 1598	62.8	5 - 9	62	63.5 ± 2.0	63.4 ± 2.1	299 ± 26	1.6	acceptable
Muck, Basal lava T765	R 1600	65.5	7 - 12	81	63.0 ± 3.4	64.1 ± 3.4	277 ± 19	1.4	acceptable
Muck, Camas Mór Gabbro	R 1597	69.8	5 - 8	46	63.1 ± 2.3	62.2 ± 2.2	314 ± 4	0.03	not acceptable

Dates have been calculated using the constants recommended in Steiger & Jager (1977) and with the age of the standard, the Bern biotite 4B, taken as 17.2 Ma.

Discussion of the new results

All the rocks sampled are reversely magnetized (or are intermediate with negative dips) except for four normally magnetized dykes from Muck. Does this imply that most of the activity was confined within a single polarity interval? The different directions for successive lava flows and the range of Curie points rule out a general remagnetization so that the reversed rocks of Muck and the main lava pile of Eigg were probably each emplaced within a single reversed interval though not necessarily the same one. However, the Sgurr pitchstone lava, which on the basis of the geological and radiometric evidence, is much younger than Rhum as well as the rest of Eigg, almost certainly represents a different reversed interval from that in which the main lava piles of both Eigg and Muck were emplaced.

The distribution of dyke mean directions is not dissimilar from other BTIP dyke collections (Mull, Ade-Hall *et al.* 1972; Skye, Wilson *et al.* 1974; Wilson, Hall & Dagley 1982; Arran, Dagley *et al.* 1978; Lundy, Mussett, Dagley & Eckford 1976; Rhum, Dagley & Mussett 1981), though there are possibly fewer shallow inclinations and certainly a much smaller percentage of normally magnetized bodies. The evidence from the lavas (Wilson *et al.* 1972b; Hall, Wilson & Dagley 1977) and igneous intrusive centres (Dagley & Mussett 1980, 1981; Dagley, Mussett & Skelhorn 1981; Hodgson 1981; Mussett, Dagley & Skelhorn 1980a; Dagley, Mussett & Skelhorn 1983) in other parts of the BTIP indicates that most of the reversely magnetized lavas and dykes formed in two relatively long intervals tentatively identified as Chron 26r (~ 2.8 Ma) and Chron 25r (~ 1.2 Ma) (Harland *et al.* 1982). This would allow ample time for the field to vary a significant amount through secular variation as has been shown by the work of Townsend & Hailwood (1981) and we believe that the dispersion of the unit mean directions for the several dyke swarms reflects such field variations.

The fact that in the lava sequences examined here successive flows (and possibly flow units) have directions of magnetization distinct from their predecessors suggests that the lavas, too, have recorded the secular variation of the magnetic field. If the flows were not extruded too rapidly, distinct directions might be expected whereas if they were intruded in quick succession, closely similar directions would result (e.g. flows 934–937 in Eigg). The directions obtained for the 'Grey Rock' lava cannot, of course, be explained in this way.

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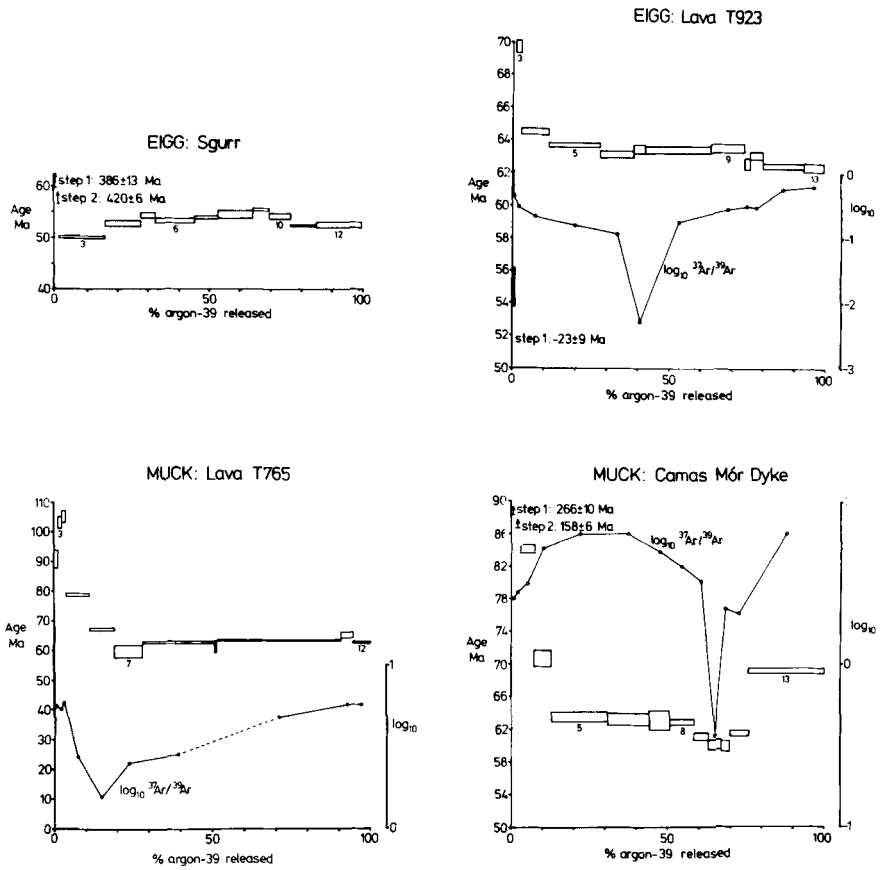


Figure 5. Age spectra of four samples. The vertical extent of each box is the age ± error (1σ), while the horizontal extent is proportional to the amount of ^{39}Ar released in that step. The circles joined by a line show the $^{37}Ar/^{39}Ar$ ratio: since ^{37}Ar derives from calcium and ^{39}Ar from potassium the ratio is a measure of the Ca/K ratio of the mineral releasing argon. No ^{39}Ar was detectable for the Sgurr sample because of the time elapsed since irradiation; however, the Ca/K ratio was probably low since the Sgurr is a pitchstone.

These new results show that

(1) The plateau lavas of Muck and Eigg and the Sgurr pitchstone lava are reversely magnetized as are two of the Eigg felsites. The third felsite and the Camas Mór Gabbro of Muck have negative but shallow inclinations which gives them intermediate polarity.

(2) The majority of the dykes are reversely magnetized, though four dykes in Muck have normal polarity (7.7 per cent of those sampled) so that there must have been at least an R-N sequence in Muck.

(3) Individual units may not always have a characteristic magnetization. This finding, which will be investigated in a separate study, makes difficult correlation of lava sequences through directions of magnetization in Muck and Eigg.

(4) There has been no general remagnetization because successive lava flows have directions of magnetization distinct from their predecessors.

(5) The lava piles have not been significantly altered by burial.

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(6) The variation of directions of magnetization in a single lava sequence and amongst the dykes is probably the result of secular variation of the magnetic field.

The relationship of Muck and Eigg to the other centres of the BTIP

All the plateau lavas within the BTIP have now been shown to be reversely magnetized (Dagley & Mussett 1981; Hall *et al.* 1977; Wilson 1970; Wilson *et al.* 1972b) but it is clear from other evidence that they are not all part of the same extrusive episode. Do they, however, belong to the same polarity interval? Using the palaeomagnetic, radiometric and geological evidence, it is possible to build up an integrated sequence (Table 1) which shows that, although most of the activity occurred within two successive polarity intervals, R, N about 59 Ma, the complete sequence spans 10 Myr and several polarity intervals.

Dykes from the Eigg swarm are truncated by gabbro bodies in Rhum (Emeleus 1973, and private communication) so that much of the Rhum central intrusive complex, the NW lavas and the late dykes which cut them, post-date the Eigg lavas. The Muck dyke swarm intersects SW Rhum (Speight 1972) so it seems likely that a similar relationship obtains between Muck and Rhum, and that the lavas of Muck and Eigg are more or less contemporary, pre-dating the Rhum Central Complex.

The lavas of NW Rhum were emplaced after a period of erosion which unroofed and produced a valley in the central complex (Emeleus & Forster 1979).

Granophyre pebbles found in a Tertiary conglomerate in Canna (Emeleus 1973) and erosional debris intercalated with the Skye lavas (Meighan *et al.* 1982) show that the Canna lavas and most of the Skye lavas, too, are younger than the Rhum intrusive centre and so are most probably contemporary with the lavas of NW Rhum.

In Skye the central intrusive complex post-dates the lavas.

Therefore, the lavas of Skye, Canna and NW Rhum were not magnetized in the same reversed interval of the magnetic field as the lavas of Eigg and Muck. The normally magnetized dykes on Rhum, Canna and probably Skye, together with the normally magnetized rocks of the Western Red Hills of Skye (Dagley & Mussett 1980) are about the same age (Dickin 1981), and would belong to the normal interval. Thus the main events in the Small Isles and Skye have recorded two polarity intervals R and N at about 59 and 58 Ma, which appear to correspond to Chrons 26r and 26 (Harland *et al.* 1982).

The Sgurr pitchstone lava in Eigg is not fed by the Grulin Felsite or the pitchstone dykes but is about 52 Myr old (Dickin & Jones 1983, and this paper) and follows a period of erosion in which valleys were scoured in the main lava pile.

On the radiometric evidence the Sgurr pitchstone and the Eastern Red Hills of Skye (Beinn an Dubhaich Epigranite, 53.5 ± 0.8 (2σ) Ma) must belong to much younger reversed intervals giving a composite sequence R-N ... R ... R for the Small Isles and Skye.

The radiometric dates are consistent with this sequence. They show that the erosional interval in Rhum must have been quite short as it is bracketed by the Western Granophyre (58.4 ± 0.8 (2σ) Ma) and a lava (60.1 ± 1.0 (2σ) Ma) (Mussett 1984). Also the dates for the Coire Uaigneich Granite, Cullin Centre (59.3 ± 1.4 (2σ) Ma) and the Loch Ainort Granite, Western Red Hills (58.7 ± 1.7 (2σ) Ma) (Dickin 1981) show that most of the igneous activity in Rhum, Canna and Skye occurred between 60 and 58 Ma within two successive polarity intervals R, N which appear to correspond to Chrons 26r and 26 (Harland *et al.* 1982).

The plateau lavas of Muck and Eigg must have been magnetized during an older reversed interval ~ 63 Ma and the Sgurr pitchstone lava during a much younger one, about 52 Ma, which may also be separate from that represented by the Eastern Red Hills (Beinn an

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Dubhaich Granite 53.5 ± 0.8 (2σ) Ma Dickin (1981); composite sill 54.9 ± 0.6 Ma Mussett, to be published).

The normally magnetized dykes found in Muck were probably emplaced before the Rhum activity and represent a different normal interval to that corresponding to the normally magnetized dykes in Rhum and Canna and the Western Red Hills of Skye. The composite sequence for the Small Isles and Skye is then R-N-R-N ... R ... R (Table 1).

We have already argued (Dagley, Mussett & Skelhorn 1984b) that the Ardnamurchan igneous centre is contemporary with that of Rhum and that the reversed interval recorded there is the same as the first reversed interval recorded by the Mull sequence, R-N-R (Mussett *et al.* 1980a, b; Dagley *et al.* 1983), which spans 61–57 Ma (Mussett, in preparation). This reversed interval therefore embraces all the activity from the lavas of Mull to the basic centres of Skye. The subsequent normal interval defines the period during which the later part of the Mull centre and the Western Red Hills of Skye were emplaced. However, the late reversely magnetized dykes cutting the Loch Ba Felsite in Mull appear to be older than either the Eastern Red Hills of Skye or the Sgurr pitchstone of Eigg. If this is so, the composite sequence for all these centres is R-N-R-N-R ... R ... R spanning 10 Myr and can be probably associated with Chrons 27r to 24r and a later reversed interval, possibly Chron 23r (Harland *et al.* 1982).

It is interesting that although lava extrusion took place at various times during this 10 Myr period, it seems always to have occurred during reversed polarity intervals.
The polarity time-scale correlations which we have discussed are only possible because of the low frequency of reversals at this time which coincides with the long reversed Chrons 26r, 25r, 24r.

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